NONSEMISIMPLE NEARBY CYCLES IN THE ANALYTIC CATEGORY

SHENG TAN

Abstract. We construct a simple irregular holonomic $\mathscr{D}$-module on $\mathbb{C}^2$ for which a graded piece of the monodromy filtration on nearby cycles is not semisimple. This gives a counterexample to the global analytic analogue of Kashiwara's conjecture and answers Problem 5.8 of Lőrincz and Yang (arXiv:2609.05215).

1. Introduction

In [Kas98], Kashiwara formulated conjectures on the semisimplicity of direct images and nearby cycles of algebraic holonomic $\mathscr{D}$-modules. The assertion (C2) concerning nearby cycles states that if Y is a smooth complex algebraic variety, g is a regular function on Y, and $\mathcal{F}$ is an algebraic holonomic $\mathscr{D}_Y$-module that is semisimple, then $\mathrm{gr}_k^{W(N_g)}\Psi_g(\mathcal{F})$ is semisimple for every integer k. Here Ψ_g denotes ordinary nearby cycles, regarded as a holonomic $\mathscr{D}_Y$-module supported on $g^{-1}(0)$, N_g is the nilpotent part of their monodromy, and $W(N_g)$ is the monodromy filtration centered at zero. Following the work on polarizable twistor modules and tame harmonic bundles [Sab05, Moc07], Mochizuki proved the algebraic conjecture in this generality using wild harmonic bundles and wild pure twistor modules [Moc11, Theorem 19.4.2].

We consider the analogue of (C2) in the analytic category, where simplicity and semisimplicity are understood globally and morphisms may have arbitrary holomorphic coefficients. The module in our example is defined algebraically and is simple in both the algebraic and analytic categories. Its algebraic nearby cycles are also simple, but their analytification is a nonsplit extension of two trivial connections.

On a complex manifold Y, the notation $\mathscr{D}_Y$ refers to holomorphic differential operators; algebraic modules are specified as such. All modules are left modules. When $D = g^{-1}(0)$ is smooth, we use $\psi_{g,1}$ for ordinary unipotent nearby cycles, regarded as a $\mathscr{D}_D$-module by Kashiwara equivalence and normalized as in §2.

2020 *Mathematics Subject Classification.* Primary 32C38; Secondary 32S40, 14F10.

Key words and phrases. Holonomic D-modules, nearby cycles, monodromy filtration, semisimplicity, Bernstein–Sato filtration.

This work is supported by NSFC (Grant No. 12601116) and BJNSF (Grant No. 1264052).

Theorem 1.1. *Let $X_{\mathrm{alg}} = \mathbb{A}^2_{\mathbb{C}}$ with coordinates x, y, let $X = X_{\mathrm{alg}}^{\mathrm{an}} = \mathbb{C}^2$, and put $f = xy - 1$ and $\theta = y\partial_y$. Define the algebraic module $\mathcal{S}_{\mathrm{alg}}$ and its analytification $\mathcal{S}$ by*

$$\mathcal{S}_{\mathrm{alg}} = \frac{\mathscr{D}_{X_{\mathrm{alg}}}}{\mathscr{D}_{X_{\mathrm{alg}}}\partial_x + \mathscr{D}_{X_{\mathrm{alg}}}(y\theta^2 - 1)}, \quad \mathcal{S} = \mathcal{S}_{\mathrm{alg}}^{\mathrm{an}}.$$

Both $\mathcal{S}_{\mathrm{alg}}$ and $\mathcal{S}$ are simple irregular holonomic modules, and multiplication by f is injective on them.

Let $D_{\mathrm{alg}} = f^{-1}(0) \subset X_{\mathrm{alg}}$ and $D = D_{\mathrm{alg}}^{\mathrm{an}}$. Identify D with $\mathbb{C}^*$ by $z = y^{-1}$, and let $E = \mathcal{O}_D u \oplus \mathcal{O}_D v$ be the connection defined by

$$z\partial_z u = v, \quad z\partial_z v = zu.$$

Then $\psi_{f,1}(\mathcal{S}) \simeq E$, the nilpotent part N_f of the monodromy on nearby cycles vanishes, and there is a nonsplit exact sequence of analytic holonomic $\mathscr{D}_D$-modules

$$0 \longrightarrow \mathcal{O}_D \longrightarrow E \longrightarrow \mathcal{O}_D \longrightarrow 0.$$

Consequently, $\mathrm{gr}_0^{W(N_f)}\psi_{f,1}(\mathcal{S}) \simeq E$ is not semisimple. The algebraic connection E_{alg} defined by the same equations on D_{alg} is simple and satisfies $\psi_{f,1}(\mathcal{S}_{\mathrm{alg}}) \simeq E_{\mathrm{alg}}$.

The connection E is associated with the classical Bessel equation $(z\partial_z)^2 F = zF$; see [XZ22, §1.1.1]. A simple irregular holonomic module may restrict to a nonsemisimple local system; an example obtained by Fourier transformation is discussed in [Sab13, p. 206]. In our construction, the irregularity along $y = 0$ forces simplicity on $\mathbb{C}^2$, while restriction to the smooth divisor $xy = 1$ gives the Bessel connection. Near this divisor, a product decomposition identifies the connection with ordinary nearby cycles. We compute the V-filtration of the graph module before and after localization, and use a Laurent series to prove that the analytic extension does not split.

The construction also answers Problem 5.8 of Lőrincz and Yang [LY26] in the analytic category. They define a Bernstein–Sato filtration $\mathcal{B}_\bullet$ on localizations of holonomic modules by imposing conditions on root multiplicities in the graph module and then evaluating the parameter. The precise definition is recalled in §3.

Corollary 1.2. *For $\mathcal{S}$ and f in Theorem 1.1, put $\mathcal{M} = \mathcal{S}_f$ and let $i : D \hookrightarrow X$ be the inclusion. For $\alpha = 0$, the Bernstein–Sato filtration satisfies*

$$\mathcal{B}_0(\mathcal{M}) = \mathcal{S}, \quad \mathcal{B}_1(\mathcal{M}) = \mathcal{M}, \quad \mathrm{gr}_1^{\mathcal{B}}(\mathcal{M}) \simeq i_+E.$$

In particular, there is a nonsplit exact sequence

$$0 \longrightarrow i_+\mathcal{O}_D \longrightarrow \mathrm{gr}_1^{\mathcal{B}}(\mathcal{M}) \longrightarrow i_+\mathcal{O}_D \longrightarrow 0.$$

In §3, we prove that dimension two is minimal for [LY26, Problem 5.8] and show that positive graded pieces of the Bernstein–Sato filtration are quotients of graded pieces of the monodromy filtration on nearby cycles. Mochizuki's theorem then gives semisimplicity of all positive graded pieces in the algebraic category.

Acknowledgements. The author thanks Ruijie Yang for pointing out the connection between an earlier version of this work and Kashiwara's conjecture on the semisimplicity of nearby cycles.

2. The analytic example

2.1. **The irregular module.** Let $X = \mathbb{C}^2$ with coordinates x, y, let $\mathscr{D}_X$ be the sheaf of holomorphic differential operators, and set $H = \{y = 0\}$ and $\theta = y\partial_y$. We consider the meromorphic connection

$$\mathcal{S} = \mathcal{O}_X(*H)e_0 \oplus \mathcal{O}_X(*H)e_1,$$

where $\mathcal{O}_X(*H)$ denotes holomorphic functions with poles of finite order along H. The differential operators act by

$$\partial_x e_0 = \partial_x e_1 = 0, \quad \theta e_0 = e_1, \quad \theta e_1 = y^{-1} e_0. \tag{1}$$

The same formulas define an algebraic connection $\mathcal{S}_{\mathrm{alg}}$ over $\mathcal{O}_{\mathbb{A}^2_{\mathbb{C}}}(*H_{\mathrm{alg}})$, where $H_{\mathrm{alg}} = \{y = 0\} \subset \mathbb{A}^2_{\mathbb{C}}$. We first identify these connections with the cyclic modules in Theorem 1.1. The section e_0 generates each connection: besides $e_1 = \theta e_0$ and $y^{-1}e_0 = \theta^2 e_0$, one has, for every integer $k \geq 0$,

$$(\theta + k)y^{-k}e_0 = y^{-k}e_1, \quad (\theta + k)y^{-k}e_1 = y^{-k-1}e_0. \tag{2}$$

Proposition 2.1. *Put $P = y\theta^2 - 1 = y^3\partial_y^2 + y^2\partial_y - 1$. In the analytic category there is an isomorphism*

$$\mathcal{S} \simeq \frac{\mathscr{D}_X}{\mathscr{D}_X\partial_x + \mathscr{D}_X P}.$$

The same presentation holds algebraically, $\mathcal{S} = \mathcal{S}^{\mathrm{an}}_{\mathrm{alg}}$, and both modules are holonomic.

Proof. Denote $C = A_1/A_1P$, where $A_1 = \mathbb{C}\langle y, \partial_y\rangle$ is the Weyl algebra in one variable. The defining operator gives the relation $y^3\xi_y^2 = 0$ on principal symbols, where ξ_y is the cotangent coordinate dual to y. Thus C is holonomic. By (2), there is a surjection from C to the meromorphic connection on the affine line. After inverting y, this is an isomorphism of rank-two connections, so its kernel is supported at zero.

The formal transpose of P is

$$P^t = y(\theta + 2)^2 - 1.$$

For every polynomial p in one variable and integer $m \geq 1$, using the relation $[\partial_y, y] = 1$, we obtain

$$\big(yp(\theta)\big)^m = \left(\prod_{j=0}^{m-1} p(\theta + j - m)\right) y^m. \tag{3}$$

Multiplication by y is locally nilpotent on a module supported at zero, so $yp(\theta)$ is locally nilpotent there as well. A quotient of A_1/A_1P^t supported at zero would have a generator q satisfying $q = y(\theta + 2)^2 q$, hence $q = 0$. Using a free resolution of C of length one, we identify its holonomic dual,

after converting right modules to left modules, with A_1/A_1P^t. By exactness of duality and biduality for holonomic modules [Sch, Lecture 8, Lemma 8.5 and Proposition 8.6], the module C has no nonzero submodule supported at zero. The kernel of the surjection vanishes.

The algebraic presentation follows by taking the external product with the trivial connection on the x-line. Exactness of analytification [HTT08, §4.7] proves the analytic presentation and the identity $\mathcal{S} = \mathcal{S}_{\mathrm{alg}}^{\mathrm{an}}$. In either category,

$$\operatorname{Char}(\mathcal{S}) \subseteq \{\xi_x = 0,\ y^3\xi_y^2 = 0\} = T^*_X X \cup T^*_H X,$$

with the same formula on the algebraic surface. Here Char is the characteristic variety, ξ_x, ξ_y are the cotangent coordinates, $T^*_X X$ is the zero section, and $T^*_H X$ is the conormal bundle of H. The right-hand side has dimension two, and the connection has rank two on $X \setminus H$. Hence the characteristic variety has dimension two. □

Proposition 2.2. *The modules $\mathcal{S}$ and $\mathcal{S}_{\mathrm{alg}}$ are simple in their respective categories.*

Proof. Let K_0 be any field of characteristic zero on which ∂_y acts trivially, and put $K = K_0((y))$. An invariant line in the connection (1) over K cannot be generated by e_0. Any other line has the form $K(he_0 + e_1)$, and invariance under θ forces

$$yh' = h^2 - y^{-1}, \tag{4}$$

where $h' = \partial_y h$. This equation has no solution in K. If $\operatorname{ord}_y h \geq 0$, the left-hand side has no negative term while the right-hand side has a nonzero term of order -1. If $n = \operatorname{ord}_y h < 0$, the two sides have orders n and $2n$, respectively; the term of order $2n$ cannot cancel y^{-1} because $2n \neq -1$. Here ord_y denotes the lowest exponent of a nonzero Laurent series.

Let $0 \neq \mathcal{T} \subseteq \mathcal{S}$ be a coherent analytic $\mathscr{D}_X$-submodule. Since $\mathcal{S}$ is torsion-free over $\mathcal{O}_X$, the submodule $\mathcal{T}$ is not supported on H. On the connected open set $X \setminus H$, it is a flat subbundle of positive constant rank. Suppose this rank is one. At a point $(a, 0) \in H$, put

$$A = \mathbb{C}\{x - a, y\}, \quad K_0 = \operatorname{Frac} \mathbb{C}\{x - a\}.$$

Expanding elements of A in powers of y gives an embedding $\operatorname{Frac}(A) \hookrightarrow K_0((y))$. The nonzero stalk of $\mathcal{T}$ generates an invariant line after this extension of coefficients, contradicting (4).

It follows that $\mathcal{T}$ has rank two and $\mathcal{R} = \mathcal{S}/\mathcal{T}$ is supported on H. If q is the image of e_0, then $q = y\theta^2 q$. By (3), $y\theta^2$ acts locally nilpotently on $\mathcal{R}$, so iteration gives $q = 0$. Since e_0 generates $\mathcal{S}$, we obtain $\mathcal{R} = 0$.

For algebraic submodules, the same argument over $\mathbb{C}(x)((y))$ excludes rank one. The quotient by a submodule of full generic rank is supported on H_{alg} and vanishes by (3). Thus $\mathcal{S}_{\mathrm{alg}}$ is simple as well. □

Proposition 2.3. *Both modules are irregular along $y = 0$.*

Proof. A regular meromorphic connection along a smooth divisor admits a logarithmic lattice; see [Sch, Lecture 25, Definition 25.4]. Restricting such a lattice to a transverse line $x = a$ and formally completing at $y = 0$ would give a finitely generated $\mathbb{C}[[y]]$-module L, stable under θ, that generates the formal connection over $\mathbb{C}((y))$. Choose an integer m such that $y^m e_0 \in L$. By (1),

$$(\theta - m)y^m e_0 = y^m e_1, \quad (\theta - m)y^m e_1 = y^{m-1} e_0.$$

Iteration gives $y^{m-j} e_0 \in L$ for every $j \geq 0$. Since L is finitely generated over $\mathbb{C}[[y]]$, the orders of its elements are bounded below, a contradiction. The same formal obstruction applies to the algebraic connection. □

We now put $f = xy - 1$, $D = \{f = 0\}$, and $\mathcal{M} = \mathcal{S}_f$. Multiplication by f is injective on $\mathcal{O}_X(*H)^2$, and the same holds algebraically. Moreover, $\mathcal{S} \subsetneq \mathcal{M}$: near any point of D, the function y is invertible and e_0/f has a simple pole.

2.2. **Nearby cycles.** The open set $U = X \setminus H$ contains the whole divisor D. We introduce coordinates on U

$$r = xy - 1, \quad z = y^{-1}, \quad x = (r + 1)z, \quad y = z^{-1}.$$

Thus $U \simeq \mathbb{C}_r \times \mathbb{C}_z^*$ and $D = \{r = 0\}$. Put $u = e_0$, $v = -e_1$, and $\vartheta = z\partial_z$. The two connections can be written as external products:

$$\mathcal{S}|_U = \mathcal{O}_{\mathbb{C}_r} \boxtimes E, \quad \mathcal{M}|_U = \mathcal{O}_{\mathbb{C}_r}(*0) \boxtimes E, \tag{5}$$

where $E = \mathcal{O}_{\mathbb{C}^*} u \oplus \mathcal{O}_{\mathbb{C}^*} v$ is defined by

$$\vartheta u = v, \quad \vartheta v = zu. \tag{6}$$

In (5), the connection in the r-direction is trivial. We identify E with a connection on D through $z = y^{-1}$, and write E_{alg} for the same connection on the algebraic divisor $D_{\mathrm{alg}} = \{xy = 1\}$.

Let $\iota : X \to X \times \mathbb{C}_t$ be the graph embedding of f. We use the decreasing Kashiwara–Malgrange filtration V^α, $\alpha \in \mathbb{C}$, with $s = -\partial_t t$ and $s + \alpha$ nilpotent on gr_V^α; see [Kas76, Mal83] and [LY26, Definition 2.1]. On $U \times \mathbb{C}_t$ set $q = r - t$ and $\eta = \partial_q$. If E_t denotes the pullback of E to the (t, z)-space, put

$$\mathscr{A} = E_t[\eta]\delta_q, \quad q\delta_q = 0.$$

This denotes finite sums of derivatives of the graph generator δ_q in the normal direction, with coefficients in E_t. In these coordinates,

$$\iota_+\mathcal{S} = \mathscr{A}, \quad \iota_+\mathcal{M} = \mathscr{A}[t^{-1}],$$

where these identities are restricted to $U \times \mathbb{C}_t$. Differentiating with respect to the original graph coordinate while keeping r fixed gives

$$\partial_{t,\mathrm{old}} = \partial_t - \eta, \quad s = -\partial_{t,\mathrm{old}} t = -(t\partial_t + 1) + t\eta. \tag{7}$$

Proposition 2.4. *On $U \times \mathbb{C}_t$, for every integer k,*

$$V^k \iota_+ \mathcal{M} = t^{k-1} \mathscr{A}, \quad V^k \iota_+ \mathcal{S} = t^{\max\{k-1,0\}} \mathscr{A}. \tag{8}$$

Both filtrations have only integral jumps. In particular,

$$\mathrm{gr}_V^0 \, \iota_+ \mathcal{S} = 0, \quad \mathrm{gr}_V^1 \, \iota_+ \mathcal{S} = \mathscr{A}/t\mathscr{A}, \quad \mathrm{gr}_V^0 \, \iota_+ \mathcal{M} = t^{-1}\mathscr{A}/\mathscr{A}. \tag{9}$$

The operators $s+1$ and s act by zero on the last two nonzero graded pieces, respectively.

Proof. The lattices for $\mathcal{M}$ are generated over $V^0 \mathscr{D}_{U \times \mathbb{C}_t}$ by $t^{k-1} u \delta_q$ and $t^{k-1} v \delta_q$, while those for $\mathcal{S}$ have the analogous generators with exponent $\max\{k-1, 0\}$. Tangential differential operators preserve these lattices, multiplication by t raises the index by one, and $\partial_{t,\mathrm{old}}$ lowers it by at most one. Moreover, the filtration of $\mathcal{M}$ is generated from the pole t^{-1} by normal derivatives and multiplication by powers of t, whereas the filtration of $\mathcal{S}$ is constant for $k \leq 1$ and is generated in higher degrees by multiplication by t. Thus both filtrations are good and exhaustive.

It remains to check the action of s on the graded pieces. On $t^{k-1}\mathscr{A}/t^k\mathscr{A}$, formula (7) gives the eigenvalue $-k$ for s, with zero nilpotent part, while the graded pieces for $\mathcal{S}$ in nonpositive degrees vanish. Together with left continuity, these properties verify the defining conditions in [LY26, Definition 2.1]. The uniqueness of the V-filtration now gives (8), from which (9) follows. □

In the convention of [Kas98, §2], the operator $t\partial_{t,\mathrm{old}} - a$ acts nilpotently on the graded piece indexed by a, and ordinary nearby cycles are defined using the indices $-1 < a \leq 0$. The index in our convention is $\alpha = a + 1$. Thus ordinary unipotent nearby cycles of $\mathcal{S}$ are represented by $\mathrm{gr}_V^1 \, \iota_+ \mathcal{S}$, with nilpotent operator $s + 1 = -t\partial_{t,\mathrm{old}}$. Its monodromy filtration agrees with that in [Kas98].

The inclusion $\iota_+ \mathcal{S} \to \iota_+ \mathcal{M}$ induces an isomorphism on gr_V^1. Multiplication by t also gives an isomorphism

$$\big(\mathrm{gr}_V^0 \, \iota_+ \mathcal{M}, s\big) \xrightarrow{t} \big(\mathrm{gr}_V^1 \, \iota_+ \mathcal{M}, s+1\big) \simeq \big(\mathrm{gr}_V^1 \, \iota_+ \mathcal{S}, s+1\big), \tag{10}$$

because $(s+1)t = ts$. Thus nearby cycles of $\mathcal{S}$ agree with the degree-zero graded piece of the localized graph module.

Let $i : D \hookrightarrow X$ be the inclusion. In $\mathscr{A}/t\mathscr{A}$, the original coordinate $r = q + t$ acts by

$$r(\eta^j e \delta_q) = -j\eta^{j-1} e \delta_q, \quad \partial_r(\eta^j e \delta_q) = \eta^{j+1} e \delta_q$$

for a local section e of E independent of t and $j \geq 0$, with the first expression zero when $j = 0$. For the direct image i_+E of a left module, these relations describe the normal direction. By Kashiwara equivalence [HTT08, Theorem 4.1.10],

$$\mathrm{gr}_V^1 \, \iota_+ \mathcal{S} \simeq i_+ E, \quad \psi_{f,1}(\mathcal{S}) \simeq E, \quad N_f = 0. \tag{11}$$

Here N_f denotes the nilpotent part of the monodromy on nearby cycles. Since the V-filtration has only integral jumps, the nearby cycles have no components with monodromy eigenvalue other than 1.

The coordinate changes and lattices above are algebraic. The same computation therefore gives $\psi_{f,1}(\mathcal{S}_{\mathrm{alg}}) \simeq E_{\mathrm{alg}}$, and identifies

$$\big(\psi_{f,1}(\mathcal{S}_{\mathrm{alg}})\big)^{\mathrm{an}} \simeq \psi_{f,1}(\mathcal{S}) \tag{12}$$

compatibly with the nilpotent endomorphisms. The algebraic connection E_{alg} is simple. Indeed, a nonzero proper algebraic submodule would be a flat subbundle of rank one. Completing it at $z = \infty$ after putting $y = z^{-1}$ and $e_0 = u$, $e_1 = -v$ would give an invariant line in (1) over $\mathbb{C}((y))$, contrary to (4).

2.3. **The nonsplit extension.** It remains to determine the analytic extension. Set

$$F(z) = \sum_{n=0}^{\infty} \frac{z^n}{(n!)^2}, \quad G(z) = \vartheta F(z). \tag{13}$$

This is the classical entire solution of the Bessel equation [XZ22, (1.1.1.1)–(1.1.1.2)]. Direct differentiation gives $\vartheta^2 F = zF$. The functions F and G have no common zero on $\mathbb{C}^*$: a common zero would give zero initial data for this second-order equation at an ordinary point, forcing $F = 0$, whereas $F(0) = 1$.

Proposition 2.5. *There is a nonsplit exact sequence of analytic holonomic $\mathscr{D}_D$-modules*

$$0 \longrightarrow \mathcal{O}_D \xrightarrow{a} E \xrightarrow{b} \mathcal{O}_D \longrightarrow 0. \tag{14}$$

Proof. Define

$$a(1) = -Gu + Fv, \quad b(u) = F, \quad b(v) = G.$$

By (6) and $\vartheta^2 F = zF$, one has $\vartheta(-Gu + Fv) = 0$, $b(\vartheta u) = \vartheta b(u)$, and $b(\vartheta v) = \vartheta b(v)$. Thus both maps are $\mathscr{D}_D$-linear. Since F and G have no common zero, the map a embeds a line subbundle, b is surjective, and $\operatorname{im} a = \ker b$.

Suppose that (14) splits. A right inverse of b sends 1 to a global horizontal section $w = A(z)u + B(z)v$ with $b(w) = 1$, where A and B are single-valued holomorphic functions on $\mathbb{C}^*$. Since w is horizontal, we have

$$A = -\vartheta B, \quad \vartheta^2 B = zB.$$

Since B is holomorphic on $\mathbb{C}^*$, it admits a convergent Laurent expansion $B = \sum_{n\in\mathbb{Z}} c_n z^n$ on a punctured disc centered at zero. The equation gives

$$n^2 c_n = c_{n-1} \quad (n \in \mathbb{Z}).$$

Taking $n = 0, -1, -2, \ldots$ shows that every negative coefficient vanishes. For $n \geq 1$, the recurrence gives $c_n = c_0/(n!)^2$. Hence $B = c_0 F$ and $A = -c_0 G$ on $\mathbb{C}^*$ by analytic continuation, so $b(w) = 0$, a contradiction. □

Both end terms in (14) are simple, so E is not semisimple. For the monodromy filtration $W(N_f)$ centered at zero, the equality $N_f = 0$ gives $W_kE = 0$ for $k < 0$ and $W_kE = E$ for $k \geq 0$. Together with (11), this gives

$$\mathrm{gr}_k^{W(N_f)}\,\psi_{f,1}(\mathcal{S}) = \begin{cases} E, & k = 0, \\ 0, & k \neq 0. \end{cases}$$

This completes the proof of Theorem 1.1.

Remark 2.6. The normal monodromy around $f = 0$, which defines $W(N_f)$, is the identity. In contrast, the tangential monodromy of E around $z = 0$ is nontrivial unipotent, as follows from the nonsplit extension (14).

The failure of semisimplicity is global: the connection E is trivial of rank two on every sufficiently small simply connected open subset of D. For $0 < \varepsilon < 1$ and $\Delta_\varepsilon = \{r \in \mathbb{C} : |r| < \varepsilon\}$,

$$f^{-1}(\Delta_\varepsilon) \simeq \Delta_\varepsilon \times \mathbb{C}^*, \quad \mathcal{S}|_{f^{-1}(\Delta_\varepsilon)} \simeq \mathcal{O}_{\Delta_\varepsilon} \boxtimes E.$$

This restriction is already nonsemisimple: a splitting of the pulled-back extension would restrict to a splitting of (14) at $r = 0$. The irregular divisor H, which forces simplicity on X, lies in $f^{-1}(-1)$.

3. Bernstein–Sato filtrations

3.1. **General properties.** Let X now be a smooth complex manifold, let $\mathscr{D}_X$ denote its sheaf of holomorphic differential operators, and let f be a nonzero holomorphic function. Let $\mathcal{S}$ be a simple holonomic $\mathscr{D}_X$-module on which multiplication by f is injective, put $\mathcal{M} = \mathcal{S}_f \neq 0$, and let $\iota : X \to X \times \mathbb{C}_t$ be the graph embedding. We follow the convention for the V-filtration in [LY26, §2.1]. For $\alpha \in \mathbb{C}$, set

$$s = -\partial_t t, \quad \Psi_\alpha = \mathrm{gr}_V^\alpha\, \iota_+\mathcal{M}, \quad N = s + \alpha.$$

The twist $\mathcal{M}f^{-\alpha}$ has the action $\xi(mf^{-\alpha}) = (\xi m - \alpha\xi(f)m/f)f^{-\alpha}$ for a local vector field ξ and a local section m of $\mathcal{M}$. In the standard presentation $\iota_+\mathcal{M} \simeq \mathcal{M}[s]f^s$, evaluation at $s = -\alpha$ defines a $\mathscr{D}_X$-linear map to $\mathcal{M}f^{-\alpha}$; see [LY26, §4.1].

For a local section w of $V^\alpha\iota_+\mathcal{M}$, let $\mu_{w,\alpha}$ be the multiplicity of $-\alpha$ as a root of its Bernstein–Sato polynomial $b_w(s)$, with the convention for the zero section in [LY26, Definition 4.8]. Lőrincz and Yang [LY26, Definition 5.1] define

$$Z_\ell V^\alpha\iota_+\mathcal{M} = \{w \in V^\alpha\iota_+\mathcal{M} : \mu_{w,\alpha} \leq \ell\},$$

and

$$\mathcal{B}_\ell(\mathcal{M}f^{-\alpha}) = \mathrm{ev}_{s=-\alpha}\big(Z_\ell V^\alpha\iota_+\mathcal{M}\big) \quad (\ell \geq 0). \tag{15}$$

Write $V^{>\alpha} = \bigcup_{\beta>\alpha} V^\beta$, with the ordering used there. By [LY26, Lemma 4.9],

$$Z_\ell V^\alpha\iota_+\mathcal{M}/V^{>\alpha}\iota_+\mathcal{M} = \ker N^\ell. \tag{16}$$

We denote by $\mathcal{S}^{-\alpha}$ the image of $V^{>\alpha}\iota_+\mathcal{M}$ under evaluation; it is the minimal extension inside $\mathcal{M}f^{-\alpha}$ described in [LY26, Proposition 4.7]. In particular, $\mathcal{B}_0 = \mathcal{S}^{-\alpha}$, as in [LY26, Proposition 5.5].

Proposition 3.1. *For every integer $\ell \geq 1$,*

$$\operatorname{gr}_\ell^{\mathcal{B}}(\mathcal{M}f^{-\alpha}) \simeq \frac{\ker N^\ell}{\ker N^{\ell-1} + N \ker N^{\ell+1}}. \tag{17}$$

If $W(N)$ is the monodromy filtration on Ψ_α centered at zero, there is a natural surjection

$$\operatorname{gr}_{\ell-1}^{W(N)} \Psi_\alpha \twoheadrightarrow \operatorname{gr}_\ell^{\mathcal{B}}(\mathcal{M}f^{-\alpha}). \tag{18}$$

Proof. By [LY26, Proposition 4.7], evaluation induces

$$\mathcal{M}f^{-\alpha}/\mathcal{S}^{-\alpha} \simeq \Psi_\alpha/N\Psi_\alpha.$$

Let $p : \Psi_\alpha \to \Psi_\alpha/N\Psi_\alpha$ be the quotient map. Equations (15) and (16) give $\mathcal{B}_\ell/\mathcal{B}_0 = p(\ker N^\ell)$. Since

$$\ker N^\ell \cap \operatorname{im} N = N \ker N^{\ell+1},$$

taking the quotient by $p(\ker N^{\ell-1})$ proves (17). The formula

$$W_k\Psi_\alpha = \sum_{j\geq 0} N^j \ker N^{k+2j+1},$$

with $\ker N^m = 0$ for $m \leq 0$, is the usual description of the monodromy filtration; see [LY26, (10)]. Its image under p is $p(\ker N^{k+1})$. Thus p maps $W_{\ell-1}$ onto $\mathcal{B}_\ell/\mathcal{B}_0$ and $W_{\ell-2}$ onto $\mathcal{B}_{\ell-1}/\mathcal{B}_0$, giving (18). □

Proposition 3.2. *For every integer $\ell > 0$,*

$$\operatorname{Supp} \operatorname{gr}_\ell^{\mathcal{B}}(\mathcal{M}f^{-\alpha}) \subseteq \{f = 0\},$$

where Supp *denotes support. If $\dim X = 1$, all positive graded pieces are semisimple.*

Proof. On $U = X \setminus \{f = 0\}$, the minimal extension $\mathcal{B}_0 = \mathcal{S}^{-\alpha}$ restricts to $\mathcal{M}f^{-\alpha}|_U$. Hence every positive graded piece restricts to zero on U. On a curve, $\{f = 0\}$ is discrete on every component on which $\mathcal{M}$ is nonzero. By Kashiwara equivalence [HTT08, Theorem 4.1.10], a holonomic module supported at one point is a finite direct sum of the simple point module. The decomposition over distinct points is locally finite in the analytic case, so every positive graded piece is semisimple. □

3.2. **The first graded piece.** We apply the definition to the surface $X = \mathbb{C}^2$ with coordinates x, y, the function $f = xy - 1$, and the module $\mathcal{S}$ of Theorem 1.1. Put $D = \{f = 0\}$, $\mathcal{M} = \mathcal{S}_f$, and let $i : D \hookrightarrow X$ be the inclusion. On $U = \{y \neq 0\}$, use $r = f$, $z = y^{-1}$, $q = r - t$, and $\eta = \partial_q$. Write E for the connection on D defined by (6), E_t for its pullback to the (t, z)-space, and $\mathscr{A} = E_t[\eta]\delta_q$ with $q\delta_q = 0$.

Proposition 3.3. *At $\alpha = 0$, the filtration satisfies*

$$\mathcal{B}_0(\mathcal{M}) = \mathcal{S}, \quad \mathcal{B}_1(\mathcal{M}) = \mathcal{M}, \quad \mathrm{gr}_1^{\mathcal{B}}(\mathcal{M}) \simeq i_+E.$$

Proof. By (8), $V^0\iota_+\mathcal{M} = t^{-1}\mathscr{A}$ and $V^{>0}\iota_+\mathcal{M} = \mathscr{A}$, while $s = 0$ on $\Psi_0 = \mathrm{gr}_V^0\,\iota_+\mathcal{M}$. Equation (16) therefore gives

$$Z_0V^0 = \mathscr{A}, \quad Z_1V^0 = t^{-1}\mathscr{A}.$$

We now use the evaluation map in (15). Since $\eta = \partial_r$ is a vector field on the original space U, its $\mathscr{D}_U$-linearity gives

$$\mathrm{ev}_0\big(\eta^j m(t,z)\delta_q\big) = \partial_r^j m(r,z)$$

for $j \geq 0$ and a local section $m(t,z)$ of $E_t[t^{-1}]$. Derivatives of coefficients holomorphic in r remain holomorphic, whereas derivatives of $r^{-1}u$ and $r^{-1}v$ generate all principal parts. Thus $\mathrm{ev}_0(\mathscr{A}) = \mathcal{S}|_U$ and $\mathrm{ev}_0(t^{-1}\mathscr{A}) = \mathcal{M}|_U$. Away from D, one already has $\mathcal{M} = \mathcal{S}$. This proves the assertions about $\mathcal{B}_0$ and $\mathcal{B}_1$.

The kernel of r on $\mathcal{M}/\mathcal{S}$ is represented by $[e/r]$, where e is a local section of E. The map from the standard model of the direct image is

$$e\delta_r \longmapsto [e/r], \quad \partial_r^j(e\delta_r) \longmapsto (-1)^j j!\,[e/r^{j+1}],$$

where $r\delta_r = 0$. It respects multiplication by r, differentiation in the normal direction, and differential operators in the tangential directions, and its image contains all principal parts. Hence $\mathcal{M}/\mathcal{S} \simeq i_+E$. □

The functor i_+ is exact and fully faithful by Kashiwara equivalence. Applying it to the nonsplit sequence (14) proves Corollary 1.2. By Proposition 3.2, this example has the smallest possible dimension for [LY26, Problem 5.8].

3.3. **The algebraic category.**

Theorem 3.4. *Let Y be a smooth complex algebraic variety, let g be a regular function on Y, and let $\mathcal{F}$ be an algebraic holonomic $\mathscr{D}_Y$-module. Assume that $\mathcal{F}$ is semisimple and that multiplication by g on $\mathcal{F}$ is injective. Then, for every $\alpha \in \mathbb{C}$ and every integer $\ell > 0$, the algebraic module*

$$\mathrm{gr}_\ell^{\mathcal{B}}(\mathcal{F}_g g^{-\alpha})$$

is semisimple.

Proof. The V-filtration, the kernels in (16), and evaluation commute with finite direct sums. Since $\mathcal{F}$ is semisimple, it suffices to consider the case where $\mathcal{F}$ is simple. Let $\iota_g : Y \to Y \times \mathbb{C}_t$ be the graph embedding, put $\mathcal{L} = \mathcal{F}_g$, and set $\Psi_\alpha = \mathrm{gr}_V^\alpha\,\iota_{g+}\mathcal{L}$ and $N = s + \alpha$, where $s = -\partial_t t$.

To identify Ψ_α with a component of ordinary nearby cycles, choose an integer n such that $\beta = \alpha + n$ satisfies $0 < \beta \leq 1$ in the chosen ordering. Since t is invertible on $\iota_{g+}\mathcal{L}$, multiplication by t^n identifies Ψ_α with $\mathrm{gr}_V^\beta\,\iota_{g+}\mathcal{L}$ and intertwines $s + \alpha$ with $s + \beta$. On the other hand, the quotient $\iota_{g+}\mathcal{L}/\iota_{g+}\mathcal{F}$ is supported on $t = 0$. In the present indexing convention, a module supported

on $t = 0$ has no positive graded pieces for the V-filtration. By exactness, we therefore obtain

$$\mathrm{gr}_V^\beta \, \iota_{g+}\mathcal{F} \simeq \mathrm{gr}_V^\beta \, \iota_{g+}\mathcal{L}.$$

With the convention of [Kas98, §2], these graded pieces correspond to a component of the ordinary nearby cycles of $\mathcal{F}$, after shifting the index by one.

We can now apply Mochizuki's theorem [Moc11, Theorem 19.4.2], which asserts the semisimplicity of the graded pieces of the monodromy filtration on algebraic nearby cycles. The ordinary nearby cycles are obtained by taking ramification degree one and zero exponential twist; see [Moc11, §22.6.3]. Consequently, $\mathrm{gr}_k^{W(N)} \Psi_\alpha$ is semisimple for every k. By (18), the module $\mathrm{gr}_\ell^{\mathcal{B}}(\mathcal{F}_g g^{-\alpha})$ is a quotient of $\mathrm{gr}_{\ell-1}^{W(N)} \Psi_\alpha$ and is therefore semisimple. □

For the module in Theorem 1.1, the formal irreducibility argument in §2 shows that its algebraic nearby cycles E_{alg} are simple. Their analytification, however, fits into the nonsplit exact sequence (14), whose maps involve the entire function F defined in (13). This function is not algebraically regular on $\mathbb{C}^*$. By (12), the analytic nearby cycles are the analytification of E_{alg}. Thus the same connection is simple in the algebraic category but not semisimple in the analytic category.

Remark 3.5. The function f can also be chosen to be the coordinate x. Let

$$X'_{\mathrm{alg}} = \operatorname{Spec} \mathbb{C}[x, y, (x+y)^{-1}], \quad X' = (X'_{\mathrm{alg}})^{\mathrm{an}}, \quad \mathcal{S}' = \mathcal{S}|_{X'}, \quad f' = x.$$

The divisor $D' = \{x = 0\} \subset X'$ is isomorphic to $\mathbb{C}^*$, and the irregular divisor $H' = \{y = 0\} \subset X'$ is nonempty. The argument of Proposition 2.2 also proves that $\mathcal{S}'$ is simple: an analytic submodule of rank one would give a formal invariant line near H', while a quotient supported on H' vanishes by (3).

Near D', we use $r = x$ as a normal coordinate and $z = y^{-1}$ as a coordinate along D'. These coordinates give the connection (6). Repeating the calculation for the graph embedding, we obtain $\psi_{f',1}(\mathcal{S}') \simeq E$, with zero nilpotent monodromy, and $\mathrm{gr}_1^{\mathcal{B}}(\mathcal{S}'_x) \simeq i'_+ E$, where $i' : D' \hookrightarrow X'$ is the inclusion and E is viewed on D' through z.

References

[HTT08] R. Hotta, K. Takeuchi, and T. Tanisaki, *D-modules, perverse sheaves, and representation theory*, Progress in Mathematics, vol. 236, Birkhäuser Boston, Boston, MA, 2008.

[Kas76] M. Kashiwara, *B-functions and holonomic systems. Rationality of roots of B-functions*, Invent. Math. **38** (1976/77), no. 1, 33–53.

[Kas98] M. Kashiwara, *Semisimple holonomic D-modules*, in *Topological field theory, primitive forms and related topics*, Progress in Mathematics, vol. 160, Birkhäuser Boston, Boston, MA, 1998, pp. 267–271.

[LY26] A. C. Lőrincz and R. Yang, *Filtrations on D-modules and multiplicities of roots of Bernstein–Sato polynomials*, arXiv:2609.05215v1 (2026).

[Mal83] B. Malgrange, *Polynômes de Bernstein–Sato et cohomologie évanescente*, Astérisque **101–102** (1983), 243–267.
[Moc07] T. Mochizuki, *Asymptotic behaviour of tame harmonic bundles and an application to pure twistor D-modules. II*, Mem. Amer. Math. Soc. **185** (2007), no. 870, xii+565 pp.
[Moc11] T. Mochizuki, *Wild harmonic bundles and wild pure twistor D-modules*, Astérisque **340** (2011), x+607 pp.
[Sab05] C. Sabbah, *Polarizable twistor $\mathcal{D}$-modules*, Astérisque **300** (2005), vi+208 pp.
[Sab13] C. Sabbah, *Théorie de Hodge et correspondance de Hitchin–Kobayashi sauvages (d'après T. Mochizuki)*, Séminaire Bourbaki, Vol. 2011/2012, Astérisque **352** (2013), Exp. No. 1050, 205–241.
[Sch] C. Schnell, *D-modules*, lecture notes, Stony Brook University.
[XZ22] D. Xu and X. Zhu, *Bessel F-isocrystals for reductive groups*, Invent. Math. **227** (2022), no. 3, 997–1092.

School of Mathematical Sciences, Capital Normal University, Beijing 100048, China

Email address: tansheng2018@outlook.com, tansheng@cnu.edu.cn